

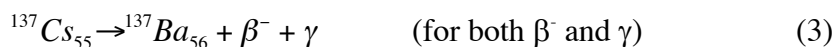
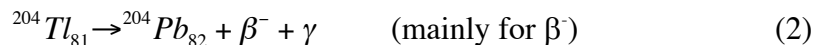
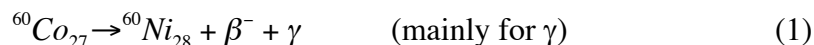
Radioactivity

1. Objectives: With all of the misconceptions and fears surrounding radioactivity, this lab's main objective is to provide you with a basic understanding of the ideas behind radioactive decay. In essence, you will learn to measure and block the particles that are the byproduct of nuclear disintegration. As part of this lab, you will learn about Geiger counters, and about the shielding of radiation.

2. Theory:

Radioactive decay is the process in which nuclei break apart, releasing energy in the process. The energy comes out in the form of particles (alpha particles and electrons), and light. Alpha particles (α) are He nuclei, made up of two neutrons and two protons solidly joined together. Electrons (or β^- particles) come from the nucleus, not the electron clouds, and they typically have very high energies. Finally, light (or γ rays) is emitted in many radioactive decays, and the wavelength of the gammas is in the region of 10^{-12} m. All of these emitted particles are called *ionizing radiation*, and they all have energies in the range 0.1-10 MeV (a MeV is one million electron volts, or 1.6×10^{-13} J). Because of their high energies they are able to ionize matter easily, thus causing biological damage when in high enough doses.

Some of the reactions we will use today include:



One final note. Radioactive decay is a natural process, happening all the time around us. At low levels of radiation, all living organisms have adapted to handle it. All of the sources we will use are very safe. They all emit $< 1 \mu\text{Ci}$ of radiation, where the unit (the Curie = Ci) is $1 \text{ Ci} = 3.7 \times 10^{10}$ decays/sec.

3. Procedure

The Geiger counter (actually Geiger-Marsden, or G-M counter) you see in the picture below consists of a controller connected to a high-voltage tube. The tube uses the ionizing property of the particles we described to count bursts in current, and therefore can measure the number of decays (within reason). Please note that you should start with the controller unplugged, and the tube connected through a coaxial cable to the controller. Your instructor can help if there are any setup problems.

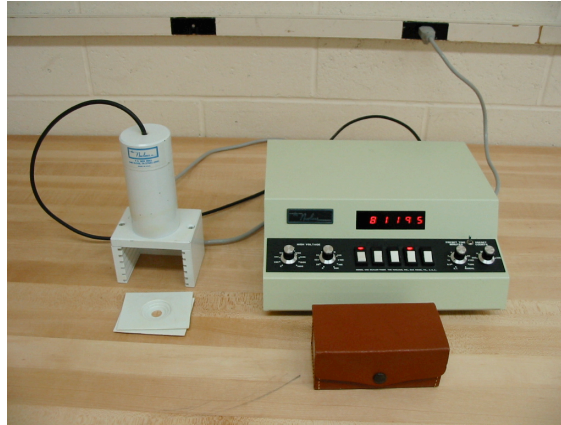


Figure 1. Setup of the Geiger counter

Operation of the Geiger counter

Below is a figure that highlights the controls of the Geiger counter. In general, once you have set the tube voltages, the operation is quite simple.

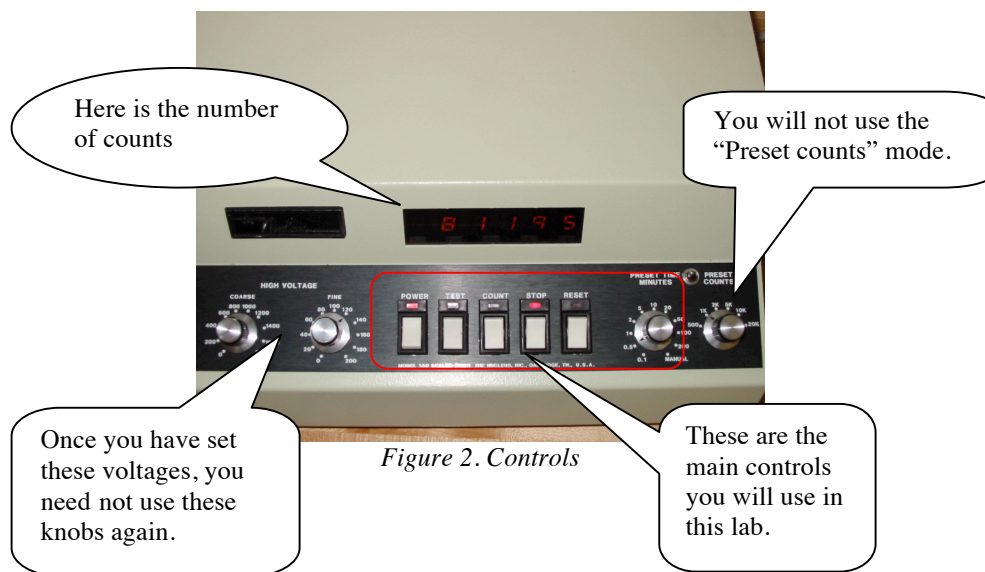


Figure 2. Controls

You turn the controller on with the Power button. The Reset. Then Count will measure decay events for a preset time (see knob at right). Once Stop light goes on, record the reading, Reset again, and you are ready for the next measurement.

Experiment 1

First: *do not be afraid!* None of the experimental materials you will handle are dangerous, unless you drop the counter on your foot. This experiment is designed to set the correct voltage for the operation of the Geiger tube. Start with the voltage knobs turned all the way to zero. Plug in the controller in to the wall outlet, and make sure that the tube is connected to the back of the controller. Set the counter to make 1-minute counts.

Now increase the voltage until you begin to get counts—taking, of course, 1 minute intervals. This is known as the threshold voltage. Once you begin to see counts, take a series of 1-minute interval counts, adding 50 volts to the operating voltage every time. You should have about 5-8 measurements. When you plot them on a graph (see Data section), you should get a graph similar to the one below. Note the quick rise (*threshold region*), and the flat part afterwards (*plateau region*). What you are looking for is a voltage in the lower part of the plateau region.

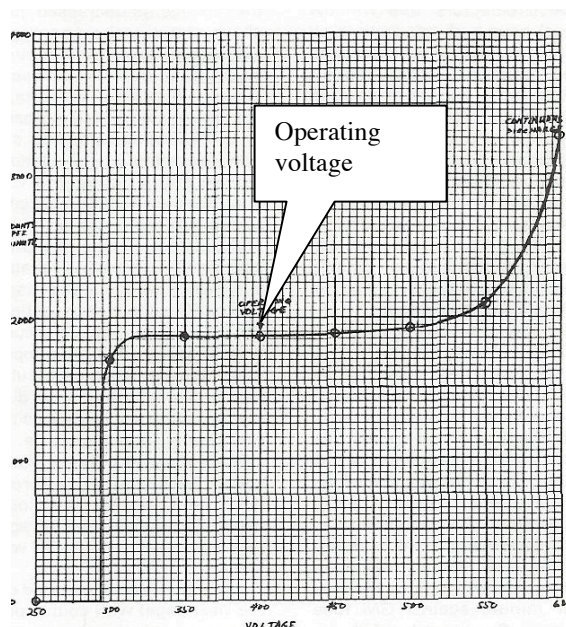


Figure 3. Characteristic curve for the GM tube.

Once you have generated a plot like the one above, choose a voltage near the bottom third of the plateau region and leave it there for the rest of the experiments.

Experiment 2

The first item is to determine the background rate. This you do by averaging the results of six 30-second intervals, with no source in the tube. To be precise, place a planchet (“shelf”) in the tube, just to keep everything constant. Make sure you have Reset the count.

Now pick one of the radioactive sources and place it on the second shelf (from the top), and take six more 30-second trials. Record the results in the Data section; you will need these results in the Analysis section.

Experiment 3

We are now going to look at the dependence of the emitted radiation with distance. One would expect an inverse square relationship (*i.e.* the number of counts goes as $1/d^2$). Count with the source on each shelf, doing four 30-second trials for each shelf. Make sure you subtract the background rate from each of your results.

Measure the distance from the source to the mica window of the G-M tube. Please be careful: do so with the apparatus off, and get the distance accurately. Also, be

careful with the window, and do not drop the tube. We shall sever one of your limbs if you do not pay attention.

Experiment 4

You will use a beta source, a gamma source, and a dual source that emits both betas and gammas. Place each source on a shelf (maybe the second or third from the top), and determine the corrected counting rate with no shielding. Do four 30-second intervals. Now shield this source with the thickest aluminum disc and again obtain the corrected counting rate.

Now repeat the procedure above for all three sources. Put the results in the data section.

Experiment 5

Use an intense source. Now get the absorber kit and measure the count rate for a range of absorbers. Try to span the entire range while doing 6-8 measurements. For each absorber, do three 30-second counts. To start, make sure the source is not too close to the tube, as some of the absorbers will be thick.

Side Note: The absorbers are described using a somewhat strange characteristic number. It is actually the product of the density times the thickness. This helps take into account possible variations in the density, which affects the absorption. Hence the units are $\text{g/cm}^3 \times \text{cm} = \text{g/cm}^2$. Do not forget, however, that the higher the number, the higher the absorption.

4. Data

Experiment 1

Plot the Counts vs. Voltage graph

Label and write down the chosen plateau voltage: _____ V

Experiment 2

Source used:

Background results: _____

Avg. (mean) **background** counts: _____

With the source: _____

Avg. (mean) **source** counts: _____

Standard deviation σ : _____

Experiment 3

Copy the background rate here (for a 30 s interval): _____ counts

Shelf positions	d (cm)	Trial 1	Trial 2	Trial 3	Trial 4	Trial Avg.
1						
2						
3						
4						
5						
6						
7						

Table 1

Experiment 4

Source		Trial 1	Trial 2	Trial 3	Trial 4	Trial Avg.
	Before Al					
	After Al					
	Before Al					
	<i>After Al</i>					
	<i>Before Al</i>					
	<i>After Al</i>					

Table 2

Experiment 5

Absorber type:_____

Absorber number	Thickness (g/cm ²)	Count 1	Count 2	Count 3	Avg. number of counts
1					
2					
3					
4					
5					
6					
7					
8					

Table 3

5. Analysis

1. What is the width of the voltage plateau? Why do we pick a lower number and not a higher one?
2. Using the data from Exp. 3, start by plotting the number of counts/sec vs. $1/d^2$, by means of the computer. Make sure you label the axes. Discuss the resulting graph. Does the data look like a straight line? If not, what do you think it trends to?
3. Discuss the results of Exp. 4 (if done). Which type of radiation was most efficiently blocked by the aluminum absorber? Do you think that this makes it more or less dangerous to living organisms?

4. Plot the data of Exp. 5, putting the count rate on the y-axis and the thickness of the x-axis. What kind of mathematical trend do you observe?

6. Questions

a. Explain the following nuclear decays, filling in what “X” is in all cases. Use your textbook if necessary:

1.

2.

3.

b. Based on your experiments, which particle (α , β , γ) has the greatest ionizing power? Explain your answer.

c. For the cases of low counts, why didn't we increase the detector voltage? Why would this be good or bad?

d. Based on your experiments, discuss the statement “Aluminum is the best radiation blocker.” Give details as to why you agree or disagree with it.